

Cultural Matching Analysis

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Overview

This analysis file examines the role of cultural matching and network opacity in predicting the persistence of social ties over time. We pool data across adjacent waves from the NetSense study to construct a person-period dataset, restricting the sample to non-kin, on-campus student ties.

We estimate the probability of tie persistence as a function of cultural matching (closed-form domains and open-ended activities) and network opacity (the number of “Don’t Know” responses regarding an alter’s preferences) using mixed-effects logistic regression (`lme4::glmer`).

Baseline Models

The following models assess the main effects of closed-form and open-ended cultural matching on the odds of tie persistence. Both closed and open-ended measures independently predict persistence.

Table 1: Odds Ratios for Tie Persistence (Baseline Models)

	Model 1: Closed	Model 2: + Open
Closed-Form Matches	1.229*** (0.060)	1.177** (0.059)
Open-Ended Matches		1.320*** (0.071)
Tie: Somewhat Close	0.514*** (0.070)	0.564*** (0.078)
Tie: Close	0.332*** (0.070)	0.369*** (0.079)
Tie Duration	0.990 (0.015)	0.991 (0.015)
Ego: Female	1.075 (0.400)	1.049 (0.393)
Alter: Female	1.400* (0.236)	1.293 (0.222)
Ego Female $\times$ Alter Female	0.628+ (0.163)	0.709 (0.187)
Tie: Same Dorm / Roommate	1.273 (0.351)	1.270 (0.353)
Tie: Race Homophily	0.970 (0.191)	0.965 (0.191)
Tie: Race Unknown	0.167*** (0.031)	0.169*** (0.031)
Period: Wave 2-5	0.050*** (0.017)	0.071*** (0.024)
Period: Wave 5-6	0.084*** (0.029)	0.113*** (0.040)
Intercept	11.509*** (5.380)	4.430** (2.225)
Num.Obs.	3251	3251
AIC	2557.8	2532.4
BIC	2643.0	2623.7
RMSE	0.32	0.32

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Network Opacity Models

These models incorporate network opacity. The main effects model reveals that a lack of shared cultural knowledge signals a weaker tie. Interaction models show that the protective effect of known cultural matches remains relatively stable regardless of how much else the ego doesn't know about the alter. Following Mize (2019), non-significant interaction coefficients are supported by formal tests of second differences in average marginal effects, which are also not statistically significant.

Marginal Effects

Figure Figure 1 visualizes the marginal effect (calculated as the change from the minimum to the maximum value) of closed-form matches, open-ended matches, and network opacity on the predicted probability of tie persistence.

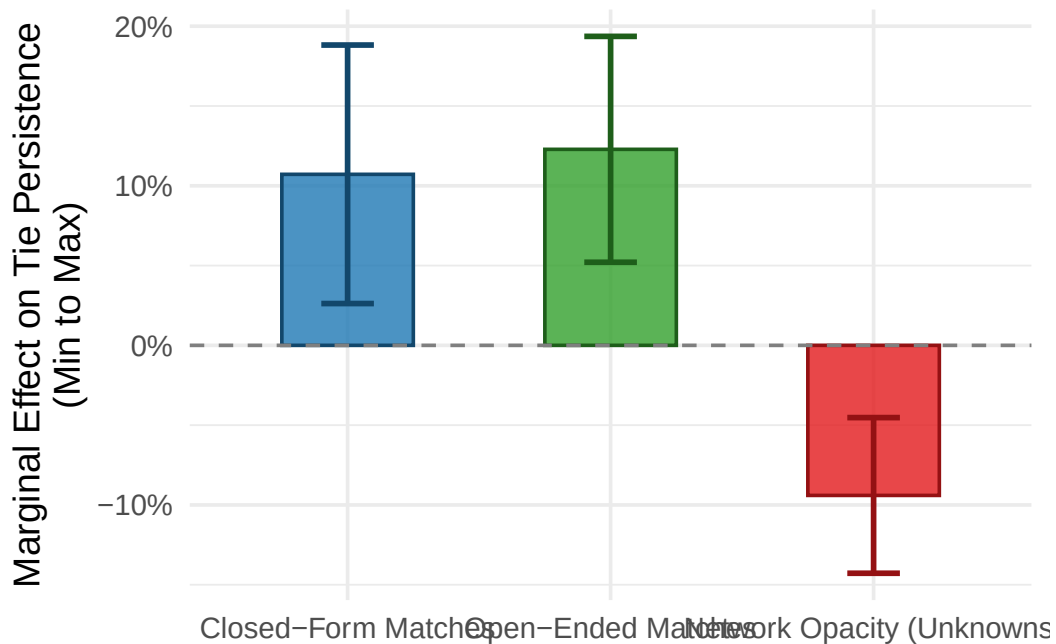


Figure 1: Marginal Effects of Cultural Matches and Network Opacity (Min to Max)

Model Fit Comparison

A comparative summary of the fit statistics for all models, demonstrating that the Opacity Main Effects model provides the best overall fit (lowest AIC/BIC) and that adding interaction terms does not statistically improve model fit.

Table 2: Odds Ratios for Tie Persistence (Network Opacity Models)

	Opacity Main Effects	Closed Interaction	Open Interaction	Both Interac
Closed-Form Matches	1.091 (0.058)	1.095 (0.060)	1.092 (0.058)	1.096+ (0.061)
Open-Ended Matches	1.291*** (0.070)	1.291*** (0.070)	1.285*** (0.076)	1.283*** (0.077)
Network Opacity (Unknowns)	0.771*** (0.045)	0.777*** (0.051)	0.761** (0.065)	0.766** (0.067)
Closed-Form $\times$ Opacity		0.989 (0.043)		0.987 (0.044)
Open-Ended $\times$ Opacity			1.007 (0.037)	
Tie: Somewhat Close	0.590*** (0.082)	0.591*** (0.082)	0.590*** (0.082)	0.591*** (0.082)
Tie: Close	0.453*** (0.099)	0.453*** (0.100)	0.452*** (0.099)	0.452*** (0.099)
Tie Duration	0.982 (0.016)	0.982 (0.016)	0.982 (0.016)	0.982 (0.016)
Ego: Female	1.035 (0.394)	1.036 (0.395)	1.033 (0.394)	1.034 (0.394)
Alter: Female	1.241 (0.215)	1.241 (0.215)	1.240 (0.215)	1.241 (0.215)
Ego Female $\times$ Alter Female	0.737 (0.196)	0.739 (0.197)	0.738 (0.196)	0.740 (0.197)
Tie: Same Dorm / Roommate	1.223 (0.344)	1.221 (0.344)	1.223 (0.344)	1.221 (0.344)
Tie: Race Homophily	0.920 (0.184)	0.920 (0.184)	0.917 (0.184)	0.916 (0.184)
Tie: Race Unknown	0.159*** (0.030)	0.159*** (0.030)	0.158*** (0.030)	0.158*** (0.030)
Period: Wave 2-5	0.062*** (0.021)	0.061*** (0.021)	0.061*** (0.021)	0.061*** (0.021)
Period: Wave 5-6	0.090*** (0.032)	0.090*** (0.032)	0.090*** (0.032)	0.090*** (0.032)
Intercept	8.084*** (4.250) ⁴	8.070*** (4.244)	8.185*** (4.333)	8.195*** (4.340)
Num.Obs.	3251	3251	3251	3251
AIC	2513.5	2515.4	2515.4	2517.3
BIC	2610.8	2618.9	2618.9	2626.9
RMSE	0.32	0.32	0.32	0.32

+p < 0.1 *p < 0.05 **p < 0.01 ***p < 0.001

Table 3: Model Fit Comparison (Log-Likelihood, AIC, and BIC)

Model	N Obs	Log Likelihood	AIC	BIC
Model 1: Closed	3251	-1264.91	2557.82	2643.03
Model 2: + Open	3251	-1251.21	2532.42	2623.72
Model 3: + Opacity	3251	-1240.73	2513.46	2610.84
Model 4: Closed Int	3251	-1240.70	2515.40	2618.87
Model 5: Open Int	3251	-1240.70	2515.40	2618.88
Model 6: Both Int	3251	-1240.66	2517.32	2626.88

Mize, Trenton D. 2019. “Best Practices for Estimating, Interpreting, and Presenting Nonlinear Interaction Effects.” *Sociological Science* 6: 81–117.